

Explosive forecasting difficulty comes from complicating the mechanics, ever so slightly. Our world, unfortunately, is far more complicated than the three body problem; it contains far more than three objects. We are dealing with what is now called a dynamical system—and the world, we will see, is a little too much of a dynamical system.

Think of the difficulty in forecasting in terms of branches growing out of a tree; at every fork we have a multiplication of new branches. To see how our intuitions about these nonlinear multiplicative effects are rather weak, consider this story about the chessboard. The inventor of the chessboard requested the following compensation: one grain of rice for the first square, two for the second, four for the third, eight, then sixteen, and so on, doubling every time, sixty-four times. The king granted this request, thinking that the inventor was asking for a pittance—but he soon realized that he was outsmarted. The amount of rice exceeded all possible grain reserves!

This multiplicative difficulty leading to the need for greater and greater precision in assumptions can be illustrated with the following simple exercise concerning the prediction of the movements of billiard balls on a table. I use the example as computed by the mathematician Michael Berry. If you know a set of basic parameters concerning the ball at rest, can compute the resistance of the table (quite elementary), and can gauge the strength of the impact, then it is rather easy to predict what would happen at the first hit. The second impact becomes more complicated, but possible; you need to be more careful about your knowledge of the initial states, and more precision is called for. The problem is that to correctly compute the ninth impact, you need to take into account the gravitational pull of someone standing next to the table (modestly, Berry's computations use a weight of less than 150 pounds). And to compute the fifty-sixth impact, every single elementary particle of the universe needs to be present in your assumptions! An electron at the edge of the universe, separated from us by 10 billion light-years, must figure in the calculations, since it exerts a meaningful effect on the outcome. Now, consider the additional burden of having to incorporate predictions about *where these variables will be in the future*. Forecasting the motion of a billiard ball on a pool table requires knowledge of the dynamics of the entire universe, down to every single atom! We can easily predict the movements of large objects like planets (though not too far into the future), but the smaller entities can be difficult to figure out—and there are so many more of them.

Note that this billiard-ball story assumes a plain and simple world; it does not even take into account these crazy social matters possibly endowed with free will. Billiard balls do not have a mind of their own. Nor does our example take into account relativity and quantum effects. Nor did we use the notion (often invoked by phonies) called the “uncertainty principle.” We are